Statistics Exercises for the 2026 Neutrino Summer School

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These exercises are a selection of homework problems from a Statistics for HEP class at UCSB.

Exercise 1

A physics analysis with a large background B seeks the extract a signal S from and extended likelihood $S+B$ fit to a multivariate discriminant. The binned PDFs for signal and background, as well as the data distributions are shown in Fig. 1. The histograms of Fig. 1 can be downloaded in `ROOT` or `npz` format from

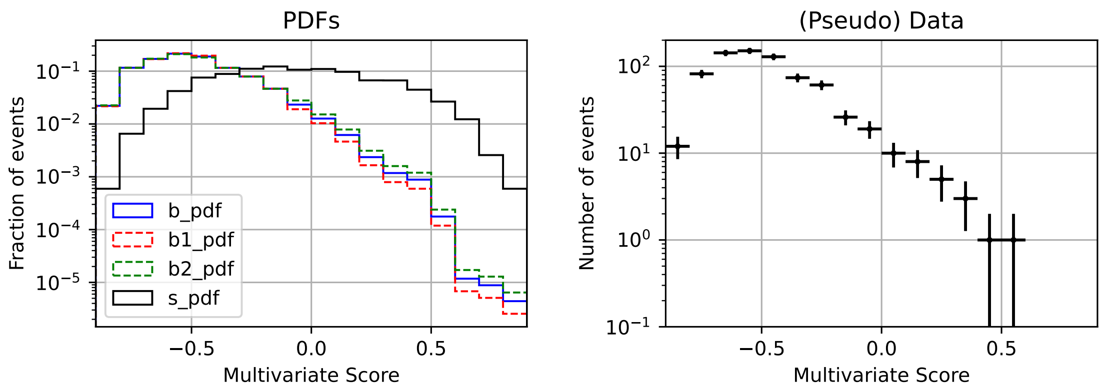


Figure 1: Left: signal and background PDFs. `b_pdf` id the background PDF, `b1_pdf` and `b2_pdf` represent the $\pm 1\sigma$ variations. Right: data)

https://tinyurl.com/3mj68xdy.

If you decide to use the `npz` file, the following code loads the histograms into `numpy` arrays:

```
# Load the histograms from npz
npzfile  = np.load("histForMinuitFit.npz")
b_pdf    = npzfile['b_pdf']
b1_pdf   = npzfile['b1_pdf']
b2_pdf   = npzfile['b2_pdf']
s_pdf    = npzfile['s_pdf']
d        = npzfile['d']         # data
binCen   = npzfile['binCen']    # bin centers
binEdges = npzfile['binEdges']  # bin edges
```

Write your own extended NLL to perform the $S+B$ fit, with and without including the background shape systematic. Use whatever approach you like to interpolate the background shape, but explain what you did. Show the $S+B$ fit result in a stacked histogram with the data superimposed. Show also a plot of the ΔNLL curve from a profile likelihood scan to S.

Hint: `Minuit` allows you to fix the value of one or more parameters. Therefore you do not need to write the log-likelohood twice, once with, and once without shape the systematic. Just write the likelihood with the systematic included, and fix the `Minuit` parameter that controls the shape to its default value when you want to turn off the systematic uncertainty.

Exercise 2

The physics of this is B-physics, not neutrino physics. But from the technical point of view it does not matter.

The angular distribution in the $D^{*+} \to D^0\pi^+$ decay can be written as a function of three complex amplitudes H_0 (longitudinal), and H_+ and H_- (transverse) as

$$\frac{dN}{d\cos\theta} \sim 4|H_0|^2\cos^2\theta + (|H_+|^2 + |H_-|^2)\sin^2\theta \tag{1}$$

where θ is the direction of flight of the D^0 in the D^* rest frame with respect to the direction of flight of the D^* in the laboratory frame. Here "longitudinal" and "transverse" refer to the three polarization states of the spin 1 D^*. The longitudinaal polarization fraction is defined as

$$\frac{\Gamma_L}{\Gamma} = \frac{|H_0|^2}{|H_0|^2 + |H_+|^2 + |H_-|^2}$$

In this folder https://tinyurl.com/bdhwmk25 there is a text file that contains 50 values of $\cos\theta$ from a toy MC using equation 1 and an experimental reconstruction efficiency constant for $\cos\theta < 0$, and then decreasing linearly to 65% of its maximum value at $\cos\theta = 1$. In other words the experimental efficiency is $\max(1, 1 - \alpha\cos\theta)\epsilon_0$ with $\alpha = 0.35$.

Write an **unbinned** log-likelihood fit to extract the longitudinal polarization factor from the MC data. (Remember: if in a NLL fit the normalizations of the PDFs depend on the fit values, the normalizations cannot be dropped). Bin the 50 toy values of $\cos\theta$ in some reasonable way, and superimpose the fitted function.

Some Physics remarks, for you to think about

1. Why is equation 1 more conveniently written as $dN/d\cos\theta$ rather than $dN/d\theta$?
2. Why is equation 1 invariant under $\theta \to \pi - \theta$, *i.e.*, $\cos\theta \to -\cos\theta$?
3. Do you understand the $\cos^2\theta$ and $\sin^2\theta$ dependencies?

Finally, the reason why experimental effects break the $\cos\theta \to -\cos\theta$ symmetry has to do with the kinematics of $D^{*+} \to D^0\pi^+$. In this decay there is very little kinetic energy liberated (look up the masses in the PDG!) so the rest-frame momentum of the decay products is small. After boosting to the lab frame the momentum of the light pion remains small. (This pion is usually called "the slow pion"). When the pion is emitted backwards with respect to the D^* flight direction ($\cos\theta > 0$) the LAB momentum of the pion is smallest, and tracking efficiency typically drops at low momentum. For an example, see Fig. 2.

Exercise 3

The experimental apparatus in the figure consists of four thin charged particle detectors at $x = 2, 3, 5$ and 7 cm. Each detector is segmented with many 50 μm- wide strips that measure the y-coordinates of charged particles going through.

Particles traveling along the y-axis in the positive x direction decay somewhere in the vicinity of ($x = 0$ $y = 0$) into two charged particles that leave hits in the detectors. You are interested in finding the x coordinate of

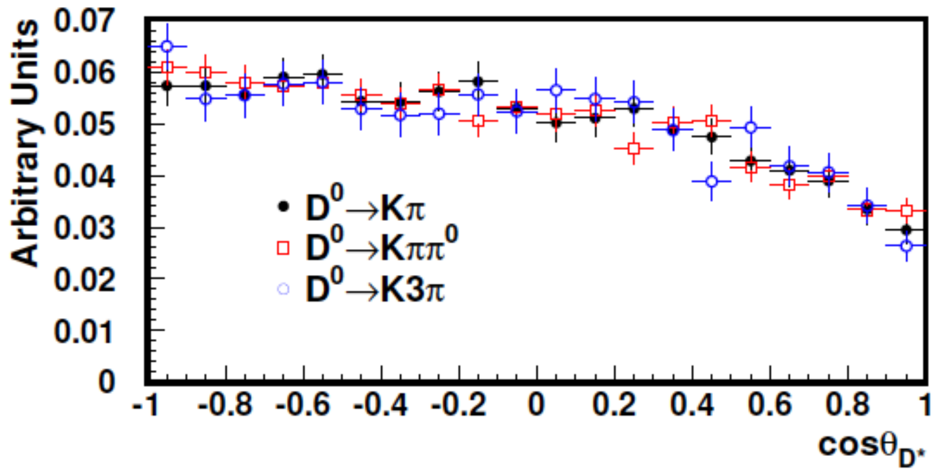
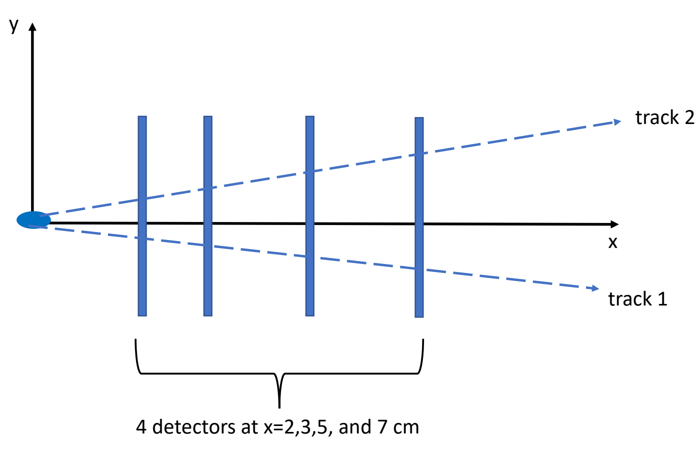


Figure 2: Efficiency for $\bar{B}^0 \to D^{*+}\omega\pi^-$ in the BaBar experiment as a function of the cosine of the decay angle in $D^{*+} \to D^0\pi^+$. From the thesis of UCSB student Bryan Dahmes.

the decay for a given pair of tracks. A simulation of 1000 such particle decays has been carried out. The simulated data are in a text file in this folder https://tinyurl.com/yc3932fd.
Each line of this file has information for one pair of tracks as follows:

```
X0 Y0 y00 y01 y02 y03 y10 y11 y12 y13
where
X0, Y0 = true common point of origin of pair (Y0 always 0)
yij = digitized coordinate of track i at detector j
All lengths in cm
```

Use whatever technique you like to **individually** fit the trajectory of the two particles. Let `Xf` be the value



of the `X` coordinate of each trajectory as it crosses the x−axis (the x-intercept).

- Plot a histogram of `Xf-X0`.
- Plot a histogram of the pull of `Xf-X0`, *i.e.*, this quantity divided by its uncertainty from the fit.
- For each pair, calculate the weighted average of the two intercepts, and plot it in a histogram.

- Plot the pull of the weighted average.

You may want to make your plots in microns rather than cm.
Note: if you use one of the common software packages for linear fits $y = Ax + B$, *e.g.*, `polyfit` in `numpy` or `TLinearFitter` in `ROOT`, remember that you will have to propagate the uncertainties of the fitted parameters A and B into the calculation of the uncertainty on the intercept.
You will find that the distribution of `Xf-X0` has very long tails, the pull distribution is fine, *i.e.*, mean ≈ 0 and standard deviation ≈ 1. This means that there is a subset of tracks for which the measurement of the intercept is not very precise, and the fit knows about it, since it reports large uncertainties in those cases.
Bonus question: Why is it that some tracks have a badly measured intercept?

For the following two exercises you need to write your own fitter using linear algebra techniques. In the same folder as the folder that contains the data for Exercise 3, you can find lecture notes from the UCSB statistics class that explain how to do it. These notes include example code in python.

Exercise 4

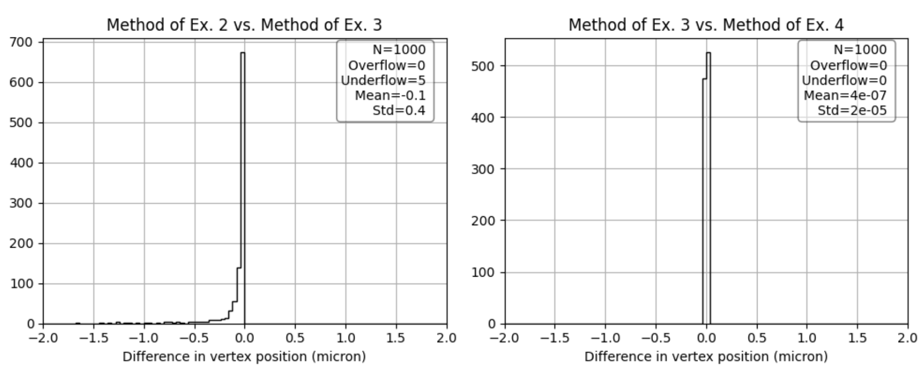
Same as Exercise 3, but now constrain the two tracks to meet at a common point `(Xf,Y=0)`. Plot a histogram of `Xf-X0` and its pull. Perform the constrained fit by eliminating one of the four parameters needed to fit the two tracks to a straight line. Instead of using some pre-packaged tool, write your own fitter using linear algebra techniques.

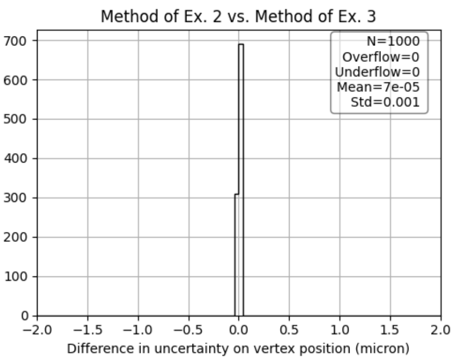
Exercise 5

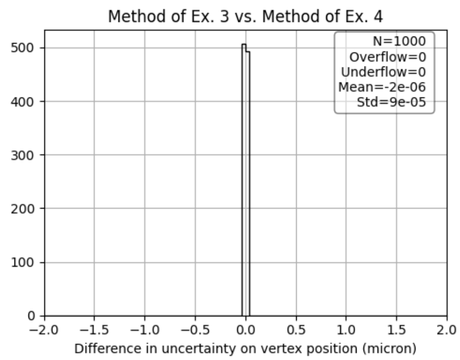
Same as Exercise 3, but now use a Lagrange Multiplier.

Comment

Here I plot the difference, track-pair by track-pair, of the results from my solutions. As expected, the constrained fits in Exercises 4 and 5 give essentially indistinguishable results. The weighted average of the vertex positions from Exercise 3 differ from the constrained fit results by typically < 1 micron. (To set the scale, the typical vertex resolution in this problem is 25 micron). Interestingly, the uncertainties on the vertex positions from all three exercises are the same. (PS: for Exercise 3 I used `polyfit`.) **Note: because I cut and pasted from the actual homeworks which had different exercise numbers, wherever the title of the plot says "Ex. n" with n=2,3,4 you should read "Ex. n+1". Sorry about that!**




Method of Ex. 2 vs. Method of Ex. 3
N=1000
Overflow=0
Underflow=0
Mean=7e-05
Std=0.001
Difference in uncertainty on vertex position (micron)


Method of Ex. 3 vs. Method of Ex. 4
N=1000
Overflow=0
Underflow=0
Mean=-2e-06
Std=9e-05
Difference in uncertainty on vertex position (micron)